Nagaraju S M

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Bengaluru, Karnataka, India - 560029

PROFESSIONAL SUMMARY

Embedded Software Engineer with hands-on experience in Firmware Over-the-Air (FOTA) embedded software testing. Experienced in Unit Testing (UT), Component Integration Testing (CIT), code-to-requirements verification, test case development. Skilled in C++ with experience in supporting FOTA systemization releases through requirements traceability and automation.

EDUCATION

- **Vellore Institute of Technology** — Vellore, Tamil Nadu, India
  Bachelor of Technology - Electronics & Communication; CGPA: 7.95 — *2020 - 2024*
- **Sarvodaya Pre-University College** — Tumakuru, Karnataka, India
  Pre-University Education - Science(PCME); Percentage: 92% — *2018 - 2020*
- **Bishop Sargant School** — Tumakuru, Karnataka, India
  Secondary Education - ICSE; Percentage: 86.6% — *2015 - 2018*

EXPERIENCE

- **Bosch Global Software Technologies** — Bengaluru, Karnataka, India (Hybrid)
  Software Engineer — *September 2026 – Present (Full-Time)*
  - Developing an end-to-end **Firmware Over-The-Air (FOTA) demonstration platform** using Python, FastAPI, WebSockets, and Raspberry Pi 5 to emulate automotive OTA update workflows. Implementing the second phase by extending the **Raspberry Pi 5-based Telematics Box Module (TBM)** to replace the vehicle TBM and interface with the existing **Secure Gateway (SGW)** and ECU environment, enabling the simulation and validation of real-world OTA firmware deployment and communication workflows.
  - Involved in Unit Testing **(UT)** and Component Integration Testing **(CIT)**. Performed **code-to-requirements** verification by executing positive and negative test scenarios, writing new test cases for newly introduced requirements, validating component interfaces and software behavior after software code changes are applied, and verifying failure handling across multiple FOTA modules.
  - Supporting **FOTA systemization** releases by improving customer requirement traceability, resolving git errors, developing Python scripts where required, and ensuring end-to-end requirements traceability and documentation delivery. Implemented a **CANoe** panel to control Network Management **(NM)** frame transmission and CAN channel frames.

Associate Software Engineer — Hyderabad, Telangana, India (On-site)
Embedded Systems & Software Testing – FOTA — *August 2025 – August 2026 (Apprenticeship)*

  - Implemented the initial phase of a Firmware Over-The-Air (FOTA) demonstration platform, including the **dashboard**, **server**, and **Git-based storage**, with VIN-based device registration, firmware campaign management, software version tracking, secure firmware repository management, and update monitoring.
  - Involved in Unit Testing **(UT)** and Component Integration Testing **(CIT)** of FOTAMaster, the core software module responsible for orchestrating firmware update sequences across multiple ECUs in **FCA (Stellantis)** vehicles. Performed **code-to-requirements** verification by executing positive and negative test scenarios, writing new test cases for newly introduced requirements, validating component interfaces and software behavior after software code changes are applied, and verifying failure handling across multiple FOTA modules.
  - Supporting **FOTA systemization** releases by improving customer requirement traceability, resolving git errors, developing Python scripts where required, and ensuring end-to-end requirements traceability and documentation delivery. Implemented a **CANoe** panel to control Network Management **(NM)** frame transmission and CAN channel frames.

- **ArkaShine Innovations** Bengaluru, Karnataka, India (On-site)
  Hardware Engineer — Founder's Office *November 2024 – August 2025 (Full-Time)*
  - Worked on Integration of portable **soil testing solutions** using Raspberry Pi and sensor integration to automate soil quality assessment and improve accessibility of soil intelligence.
  - Led deployment initiatives across **Hyderabad & Wanaparthy (Telangana)**, **Raichur (Karnataka)**, and **Wayanad (Kerala)**, overseeing on-site device calibration, field testing, and remote troubleshooting to improve system reliability.
  - Contributed to **microbial activity** analysis by standardizing soil gas emission measurements through CO_2-equivalent conversion methods.
  - Designed a revenue model for a **Micro-Entrepreneur Business** approach, enabling employment opportunities while making soil intelligence solutions more affordable to smallholder farmers. Collaborated closely with the Co-founder on key engineering design decisions, **deployment strategy**, and **product refinement**, contributing to key design and operational decisions.

INTERNSHIP

- **Bosch Global Software Technologies** Coimbatore, Tamil Nadu, India (On-site)
  ASE Project Trainee *Jan 2024 – May 2024*
  - Supported the COM-STEERING project by managing software requirements in DOORS and ensuring alignment with customer requirements and software development activities.
  - Worked with automotive development and analysis tools including CANalyzer, Trace32, Enterprise Architect, Astree, and SystemDesk, supporting project activities and technical documentation.

PROJECTS

32-DOF Humanoid Robotics Main Processing & Communication Unit *Python*

- Designed and presented a main processing and communication unit for a 32-DOF humanoid robot using Webots, with a Python-based GUI for motion control and smooth behavior generation.

Smart Manufacturing:
Fuel Efficiency and Turbine Health Analytics on Azure Cloud *Azure, Python, FastAPI*

- Developed an Azure-based analytics pipeline to monitor turbine performance, fuel-efficiency trends, and health indicators using historical operational data, supporting anomaly detection and condition monitoring.

ACHIEVEMENTS & CERTIFICATIONS

- **Hackathon Winner**: FIT.Fest'25, BOSCH
- **Tech Quiz Finalist**: FIT.Fest'25, BOSCH
- **Microcontroller Embedded C Programming**: Udemy
- **Young Professional Career Edge**: TCS iON

SKILLS

Programming:	C, C++, Python, SQL, CAPL
Automotive:	FOTA, CAN, CAN FD, Requirements Analysis
Testing:	VectorCAST, CANoe, CANalyzer, Trace32, CIT, UT
Tools:	Git, GitHub, DOORS, FastAPI, Raspberry Pi OS, Eclipse CDT
Embedded Platforms:	8051, Arduino Uno, Raspberry Pi 4 Model B, Raspberry Pi Pico W